

Module 06: Learning — Synaptic Plasticity as Model Updating

Course: Active Inference 401 | **Unit:** Neuroscientific Frontiers | **Audience:**
Advanced undergraduates / graduate students

Learning Objectives

1. Analyze how **synaptic plasticity** (LTP, LTD, STDP) implements parameter learning — updating the generative model based on experience.
2. Evaluate the evidence for **Bayesian model reduction** in the brain — sleep, pruning, and structure learning.
3. Examine **memory consolidation** as the transfer of learned parameters between brain systems.

Key Concepts

1. Synaptic Plasticity as Parameter Learning

In Active Inference, learning is the updating of generative model parameters — the A, B, C, D matrices. Synaptic plasticity is the neural mechanism:

Long-term potentiation (LTP): Strengthening of synaptic connections when pre- and post-synaptic neurons are co-active. In Active Inference terms, LTP implements the strengthening of associations in the generative model — when two states frequently co-occur, the A-matrix entry linking them increases. This is the neural implementation of Dirichlet concentration parameter accumulation.

Long-term depression (LTD): Weakening of synaptic connections. This implements the reduction of unlikely associations — when predicted co-occurrences fail to materialize, the corresponding model parameters decrease. In the cerebellum, LTD at parallel fiber-Purkinje synapses updates the forward model based on motor prediction errors.

Spike-timing-dependent plasticity (STDP): The precise timing of pre- and post-synaptic spikes determines whether LTP or LTD occurs. If pre fires before post (causal), LTP occurs — strengthening the model's causal prediction ("this input predicts this output"). If post fires before pre (acausal), LTD occurs — weakening the incorrect causal link. STDP implements **causal inference** at the synaptic level — learning the temporal structure of the world.

2. Prediction Error-Driven Learning

Learning in Active Inference is driven by prediction errors — the discrepancy between what the model predicts and what occurs. Several neural learning rules implement this:

Hebbian learning with a predictive twist: Classical Hebb: "Neurons that fire together wire together." Predictive refinement: "Neurons that *predict* each other strengthen their connection; neurons that *fail to predict* each other weaken their connection." The prediction error modulates the learning signal — large errors produce large updates, small errors produce small updates.

Reward prediction errors (RPE): Phasic dopamine signals from VTA/SNc encode reward prediction errors — the difference between expected and received reward. These RPEs update value estimates in the striatum and PFC, refining the C vector (preferences) and policy evaluations. This is the neural mechanism of preference learning.

Sensory prediction errors: Cortical prediction errors (carried by feedforward gamma) drive learning in the sensory hierarchy — updating the A-matrix (observation model). The precision of the error signal determines the learning rate: high-precision errors (salient, reliable) produce faster learning; low-precision errors (noisy, unreliable) produce slower learning.

3. Memory Consolidation and Systems Learning

Memory consolidation — the transfer of newly learned information from hippocampus to neocortex — is a critical component of generative model updating:

Hippocampal-cortical dialogue: During waking experience, the hippocampus rapidly encodes new observations (fast learning rate, high plasticity). During sleep, the hippocampus replays these experiences to the neocortex, which slowly integrates them into its generative

model (slow learning rate, gradual parameter updating). This implements a **complementary learning systems** architecture.

Sharp wave-ripples (SWR): During NREM sleep, hippocampal sharp wave-ripples replay compressed versions of waking experiences. These replays are coordinated with neocortical slow oscillations (< 1 Hz) and thalamocortical spindles (12-15 Hz), creating a temporal window for cortical plasticity. This three-way coupling (SWR $\rightarrow$ spindle $\rightarrow$ slow oscillation) is the neural mechanism of consolidation.

Sleep spindles: Thalamocortical spindles (~ 12 -15 Hz) trigger cortical calcium influx that facilitates LTP. This is the plasticity mechanism through which replayed memories are "written" into cortical synapses. Individual differences in spindle density predict learning ability.

4. Bayesian Model Reduction and Synaptic Pruning

Bayesian Model Reduction (BMR) — simplifying the generative model by removing unnecessary parameters — has a neural counterpart in **synaptic pruning**:

Developmental pruning: The adolescent brain undergoes massive synaptic elimination, reducing the number of synapses by $\sim 40\%$. This is not random loss — it is the removal of synapses that contribute little to model evidence. The cortical generative model becomes more efficient by eliminating unnecessary parameters.

Sleep-dependent pruning: The **synaptic homeostasis hypothesis** (Tononi & Cirelli) proposes that sleep involves global synaptic downscaling — weakening all synapses while preserving relative differences. Active Inference interpretation: sleep implements a form of BMR, pruning weak connections (low-evidence parameters) while maintaining strong ones (high-evidence parameters). This explains why sleep improves generalization — the pruned model overfits less.

Reconsolidation: When a memory is reactivated, it temporarily becomes labile (subject to updating or erasure). Active Inference: reactivation opens a window for model revision — the retrieved parameters can be updated in light of new evidence. This has clinical implications for treating PTSD (reconsolidation-based interventions).

5. Neurogenesis and Structure Learning

Adult hippocampal neurogenesis — the birth of new neurons in the dentate gyrus — may implement **structure learning** — adding new nodes to the generative model:

New neurons for new contexts: Newly born neurons are hyperexcitable and hyper-plastic, making them ideal for encoding novel information. They may create new "slots" in the hippocampal generative model for representing new contexts or environments.

Pattern separation: New neurons improve the hippocampus's ability to discriminate between similar contexts (pattern separation), reducing interference between overlapping generative model representations.

Environmental enrichment: Enriched environments increase neurogenesis, consistent with the idea that structure learning is driven by environmental complexity — new experiences require new model structure.

6. Meta-Learning and Learning Rate Optimization

Active Inference includes a mechanism for **learning to learn** — optimizing the learning rate itself:

Dopaminergic precision modulation: Tonic dopamine levels modulate the precision of prediction errors across the cortical hierarchy. Higher tonic dopamine → higher precision → faster learning from prediction errors. Lower tonic dopamine → lower precision → slower, more conservative updating. This implements a **meta-learning** signal that adjusts the organism's overall sensitivity to new evidence.

Context-dependent learning rates: The hippocampus detects environmental change (context shifts) and signals the need for rapid model updating. When the hippocampus detects a novel context (mismatch between current observations and stored representations), it increases learning rates across the cortical hierarchy — enabling rapid adaptation to new environments.

Cholinergic modulation: Acetylcholine from the basal forebrain increases the precision of sensory prediction errors relative to prior predictions. High ACh → data-driven learning (the model trusts observations over priors). Low ACh → prior-driven inference (the model trusts

its existing model). This implements the explore/exploit dimension of learning: curiosity states (high ACh) versus consolidation states (low ACh).

Clinical Vignette — Alzheimer's Disease and Learning Failure: Early Alzheimer's selectively damages cholinergic neurons in the basal forebrain and hippocampal circuits. In Active Inference terms, this disrupts both the precision modulation of prediction errors (cholinergic loss) and the rapid encoding of new experiences (hippocampal atrophy). The result: patients cannot update their generative model — they repeat the same predictions despite mounting prediction errors. The characteristic confabulation of Alzheimer's represents a model that can no longer learn, generating predictions from an increasingly outdated generative model. Cholinesterase inhibitors partially restore learning by boosting residual cholinergic signaling.

Cross-Track Connection — Advanced Theory (Module 06): The meta-learning mechanisms here implement the hierarchical Dirichlet process described formally in the Advanced Theory track — learning rates themselves are hyperparameters with their own precision, creating a hierarchy of learning.

Summary

Synaptic plasticity (LTP, LTD, STDP) implements parameter learning in the generative model. Prediction errors drive learning, with different types (sensory, motor, reward) updating different model components. Memory consolidation transfers parameters from hippocampus to neocortex through sleep-dependent replay. Synaptic pruning and sleep-dependent downscaling implement Bayesian Model Reduction. Meta-learning — optimizing the learning rate through dopaminergic and cholinergic modulation — enables adaptive sensitivity to environmental change. Adult neurogenesis may support structure learning — adding new representational capacity to the model.

Further Reading

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- Tononi, G. & Cirelli, C. (2014). Sleep and the price of plasticity. *Neuron*, 81(1), 12-34.
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- Yu, A. J. & Dayan, P. (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681-692.
- Hasselmo, M. E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology*, 16(6), 710-715.